

LECTURER: Nghia Duong-Trung

DEEP LEARNING

Introduction to Neural Networks and Deep Learning

1

Network Architectures

2

Neural Network Training

3

Alternative Training Methods

4

Further Network Architectures

5

Master Thesis Topics

x.1

UNIT x.1

Master Thesis Topics on Recent Trends in AI

1. AGENTIC MULTI-AGENT SYSTEMS FOR COMPLEX TASK AUTOMATION

Idea: Build and evaluate a multi-agent system (e.g. research assistant team or data pipeline engineer + QA agent) using an open-source agentic framework.

Prototype: Use AG2 (formerly AutoGen): github.com/ag2ai/ag2 – to implement multi-agent coordination, tool use, and human-in-the-loop workflows.

Research questions:

- How do different coordination patterns (manager-worker vs. peer-to-peer) affect task success rate, latency, and cost?
- Can we design evaluation benchmarks that disentangle single-model capability from agentic orchestration quality?
- How robust is an agent team to tool failures or inconsistent model outputs?

2. SMALL LANGUAGE MODELS (SLMS) IN AGENTIC PIPELINES

Idea: Investigate SLMs as efficient alternatives to large LLMs inside agentic workflows, focusing on cost, energy, and deployment constraints.

Prototype: Select a few open SLMs (<10B parameters), apply quantization/pruning libraries, and integrate them into a simple multi-tool agent pipeline.

- <https://huggingface.co/blog/jjokah/small-language-model>

Research questions:

- For which tasks do SLMs match or surpass LLMs when wrapped in an agentic workflow?
- How do compression techniques (quantization, pruning) affect downstream agent behavior (tool calls, error rates, hallucination)?
- What are the trade-offs between quality, latency, and energy usage for SLM- vs. LLM-centered agents?

3. LIGHTWEIGHT RAG WITH GRAPH-ENHANCED INDEXING

Idea: Move from simple vector search to graph-augmented retrieval where documents, sections, and concepts are connected as a graph.

Prototype: Use LightRAG: github.com/HKUDS/LightRAG – to build a graph-aware RAG system with a small multi-hop QA demo.

Research questions:

- How does graph-enhanced RAG compare to vanilla dense RAG on multi-hop QA and multi-document reasoning tasks?
- How do different graph structures (concept graph, citation graph, section graph) influence retrieval quality and hallucination rate?
- Can we design a budget-aware RAG that trades off retrieval depth and graph exploration against latency and cost?

4. EDGE AI WITH VISION TRANSFORMERS (ViT) AND TINYML

Idea: Deploy Vision Transformers (e.g. EfficientViT) on edge devices and compare them to lightweight CNNs for real-time vision applications.

Prototype: Implement a real-time detection or classification task (traffic signs, classroom occupancy, safety monitoring) on a Raspberry Pi or similar device.

- <https://hanlab.mit.edu/projects/tinyml>

Research questions:

- What is the best trade-off between model size, frame rate, and detection accuracy under different hardware constraints?
- Do transformer-based models still outperform CNNs once heavily quantized and pruned on low-power devices?
- How do different optimization strategies (quantization, pruning, distillation) impact robustness in real-world edge scenarios?

5. LIGHTWEIGHT CNNs FOR ON-DEVICE PERSONALIZATION

Idea: Train a generic CNN model centrally, but allow **small personal adaptations** at the edge (few-shot personalization).

Example:

- Face/gesture recognition for a single user (or a household).
- Generic CNN backbone + tiny user-specific adaptation layers or LoRA-style adapters.
- Study few-shot learning performance, privacy, and computational cost.

6. OMNIVINCI: ENHANCING ARCHITECTURE AND DATA FOR OMNI-MODAL UNDERSTANDING LLM

Idea: The ability to perceive across multiple modalities.

- <https://github.com/NVlabs/OmniVinci>
- <https://nvlabs.github.io/OmniVinci/>

7. BUILD REASONING LLMS USING GRPO

Idea: Fine-tune a base model by using a reinforcement learning algorithm to optimize its reasoning abilities based on a reward function, typically through a process that involves a dataset, system prompt, multiple candidate responses, and a loss function for gradient updates.

- <https://docs.unsloth.ai/get-started/reinforcement-learning-rl-guide/tutorial-train-your-own-reasoning-model-with-grpo>
- <https://lightning.ai/lightning-purchase-test/environments/build-a-reasoning-llm-from-scratch-using-grpo?section=featured>